KIRAN UPADHYAY

AI & Agentic AI Engineer | Python | LLM Applications | RAG | Backend APIs | Evaluation & Automation

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AI and GenAI professional with 3+ years of experience in Python-based LLM applications, retrieval systems, AI workflow automation, and model evaluation. Developed GraphFix, an MIT-licensed software issue localization system evaluated across 560 benchmark issues, and Budget-Aware AutoLLM, a reasoning framework with 379 automated tests. Experienced in deterministic validation, GitHub Actions CI, API integrations, reproducible experimentation, and evidence-based AI workflows. Currently pursuing an M.Tech in Data Science and Artificial Intelligence at PES University.

TECHNICAL SKILLS

AI & Agentic Engineering: Large Language Models, AI Agents, LLM Orchestration, Retrieval-Augmented Generation (RAG), Agentic Workflows, Prompt Engineering, Structured Evaluation, QLoRA, Fine-tuning

Python & Backend Engineering: Python, FastAPI, REST APIs, API Integration, SQL, SQLAlchemy, Workflow Automation, Pandas, NumPy

Retrieval & Data Processing: LangChain, FAISS, BM25, Dense Retrieval, Hybrid RRF, Data Processing, Hugging Face, PyTorch, TensorFlow

Testing & Reliability: Pytest, Automated Testing, Regression Testing, Deterministic Evaluation, LLM Evaluation, Benchmarking, Failure Analysis, Reproducible Experiments, Ruff, mypy

Development & Deployment: Git, GitHub, GitHub Actions, CI/CD Workflows, Docker, Streamlit, Jupyter Notebook, Ollama, OpenAI API

RESEARCH & PUBLICATION

When Is More Reasoning Worth It? Fallback-Aware Reasoning-Strategy Routing Under Hard Token Budgets

Kiran Upadhyay (Sole Author) — SSRN Preprint (Elsevier), October 2026

DOI: 10.2139/ssrn.7544241

- Studied fallback-aware selection among reasoning strategies under hard token budgets using frozen routing policies and conservative resource prediction.
- Evaluated Qwen2.5-3B-Instruct and Phi-3.5-mini-instruct across four reasoning benchmarks with 2,400 routed and 4,000 exhaustive executions; reduced macro budget-violation rate from 0.258 to 0.033 (Qwen) and 0.225 to 0.038 (Phi).

PROFESSIONAL EXPERIENCE

Pareto AI | Generative AI Prompt Engineer | Freelancer | Aug 2024 – Oct 2026

- Developed and evaluated Python-based LLM workflows for document intelligence, mathematical reasoning, information retrieval, and business-process automation.
- Built reusable evaluation pipelines for reasoning accuracy, model reliability, hallucination analysis, and failure identification.
- Implemented Python automation for workflow orchestration, API integrations, repeatable model testing, and structured results analysis.

Outlier | Generative AI Prompt Engineer | Freelancer | Aug 2023 – Aug 2024

- Designed mathematical reasoning tasks, logical problem-solving scenarios, and evaluation benchmarks for Large Language Models.
- Evaluated outputs for mathematical correctness, reasoning consistency, hallucinations, and response quality; built LLM-powered search agents integrating external APIs.

Mathematics Subject Matter Expert | Sep 2020 – Oct 2023

Chegg India Pvt. Ltd. | Straive | Dimentics Info Tech Pvt. Ltd. | Scholars Wing Edu Tech Solutions LLP | Unacademy

- Developed 5,000+ mathematically rigorous solutions across Calculus, Linear Algebra, Probability, Statistics, Optimization, Numerical Methods, and Abstract Mathematics.

OPEN SOURCE CONTRIBUTIONS

- **Statsmodels**: Implemented heterogeneous-scale t-test power calculations with numerical validation and regression tests (PR #10326, open).
- **Hugging Face Optimum**: Added PLBart configuration support with focused regression testing and style checks (PR #2502, open).
- **Flet**: Fixed Gallery links and implemented permanent redirects after the Flet Studio migration (PR #6770, merged).
- **Hugging Face Hub**: Translated inference documentation into Hindi while preserving technical references (PR #4963, open).

SELECTED AI & GENAI PROJECTS

GraphFix – Graph-Guided AI Agent for Software Issue Localization | GitHub

- Developed a modular Python-based software issue localization system combining AST parsing, repository graphs, BM25, MiniLM dense retrieval, Hybrid RRF, and bounded adaptive graph navigation.
- Evaluated 560 Loc-Bench V1 issues using exact pre-fix repository snapshots and deterministic evaluation, achieving zero evaluation failures.
- Improved file-level Hit@1 from 30.0% to 39.5% and Mean Reciprocal Rank from 0.415 to 0.502 compared with the BM25 baseline.
- Published an MIT-licensed Python project with GitHub Actions CI, 88 passing tests, Ruff, strict mypy, and a Streamlit live-repository explorer.

Budget-Aware AutoLLM | GitHub

- Developed an open-source Python framework for budget-aware LLM reasoning-strategy routing with deterministic experiments, reproducible reporting, and Streamlit-based analysis.
- Implemented five reasoning strategies and four routing policies across token, latency, and optional estimated-cost constraints.
- Established a GitHub Actions CI workflow with 379 deterministic automated tests to support regression validation and reproducible research.

Agentic AI Evidence-Based Clinical Research Assistant | GitHub | Hugging Face

- Developed a Python-based biomedical RAG system integrating BioMistral-7B, PubMedBERT embeddings, FAISS vector search, and QLoRA for evidence-based clinical research.
- Built an end-to-end workflow covering document preprocessing, semantic indexing, query understanding, retrieval planning, evidence selection, source-cited generation, and response validation.
- Processed 152 biomedical articles into 2,267 searchable chunks and developed a Streamlit application for evidence inspection, workflow tracing, and retrieval evaluation.

Maria – AI Voice Agent for Banking Collections

- Built an AI-powered conversational voice agent using Vapi and OpenAI APIs with Speech-to-Text, Text-to-Speech, LLM orchestration, behavioral guardrails, and workflow automation.

MACHINE LEARNING ACHIEVEMENTS

- 1st Rank – Bengaluru House Price Prediction
- 1st Rank – Fake vs. Real Image Detection
- Top 5 – Employee Attrition & Sunspot Forecasting
- Top 7 – NLP Sentiment Analysis

EDUCATION

PES University | M.Tech. Data Science & Artificial Intelligence | Expected Dec 2026

CGPA: 9.81/10

Marwadi University | M.Sc. Mathematics | 2018

CGPA: 9.55/10

Saurashtra University | B.Sc. Mathematics | 2016

CGPA: 8.65/10